Effects of Dissolved Oxygen Levels on the Greenhouse Gas Emissions and Denitrification Performance of Woodchip Bioreactors Treating Onsite Wastewater

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Introduction

Nitrogen Removal from Wastewater

- Nitrogenous waste in the environment leads to eutrophication and the proliferation of harmful algal blooms along with diseases, threatening biodiversity and human health (Diaz-Elsayed *et al.*, 2017; Salo and Salovius-Laurén, 2022; Gobler *et al.*, 2021; Ward *et al.*, 2018)
- **Onsite wastewater treatment systems** make up 25% of total wastewater treatment in the US, serving households without access to municipal wastewater treatment (Lee *et al.*, 2021)

Types of Onsite Wastewater Treatment

Conventional onsite wastewater treatment systems (Figure 1) remove 10-40% of total nitrogen (Diaz-Elsayed *et al.*, 2017)

- Done by naturally occurring microbes in soil
- Oxygen depletion in soil hinders aerobic nitrifiers (Cox *et al.*, 2020)

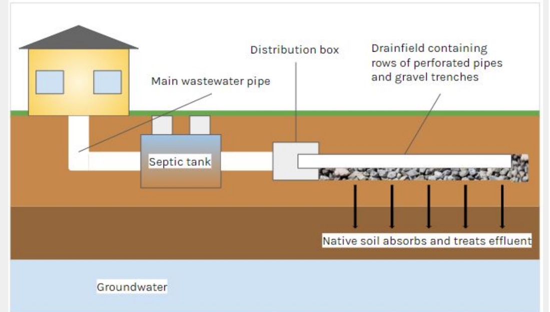


Figure 1. Diagram of conventional septic system for onsite wastewater treatment. (Image generated by student)

Nitrogen-removing biofilters (NRBs) (Figure 2) achieve 80-98% removal (Chen *et al.*, 2022)

- Done by creating separate conditions for nitrification and denitrification to maximize the effectiveness of each process (Figure 3)

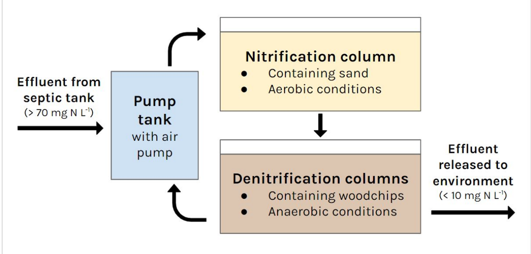


Figure 2. Diagram of FlexTreat® biofilter (Chen et al., 2022). (Image generated by student)

- **This study focuses on NRBs as they remove nitrogen more effectively, holding more promise in future use.**

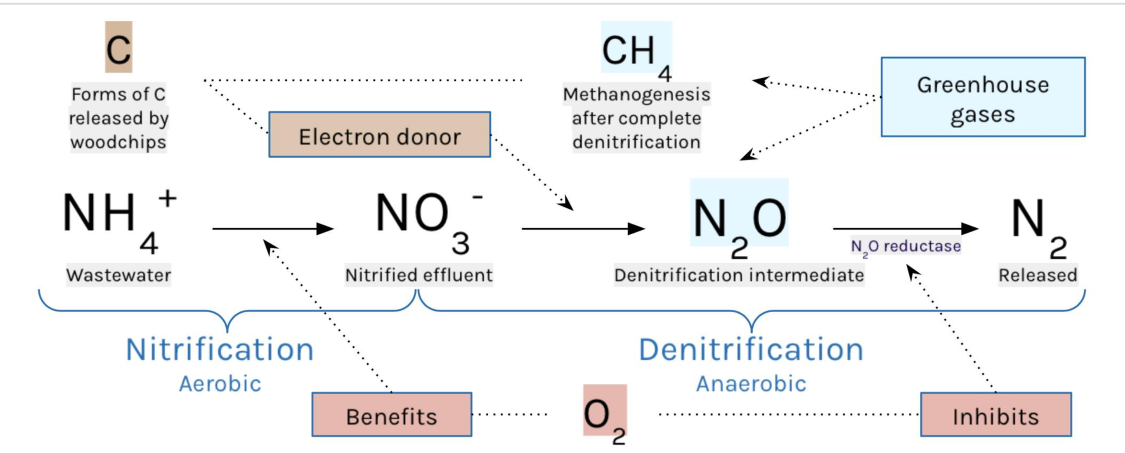


Figure 3. Process of nitrogen removal by NRBs, including greenhouse gases and dissolved oxygen. (Image generated by student)

Role of Dissolved Oxygen (DO)

- **Nitrification:** DO >2 mg L^{-1} (Beccari *et al.*, 1992)
- **Denitrification:** Incomplete when DO = 3.3-5.8 mg L^{-1} (Chen *et al.*, 2022)
  - NO_3^- used as terminal electron acceptor only when oxygen is limited (Maleki Shahraki *et al.*, 2021)
  - Expression of nosZ gene encoding N_2O reductase is sensitive to oxygen (Spiro, 2012)
- **Effects of altering oxygen availability in NRBs have yet to be investigated**

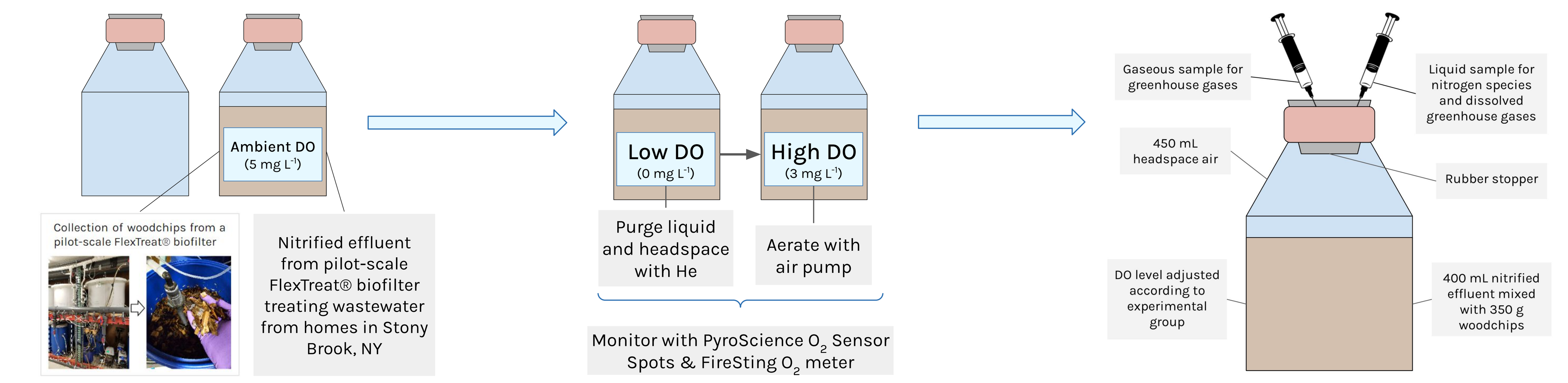
Greenhouse Gas Emissions

- Wastewater treatment results in the production of critical greenhouse gases (Davis *et al.*, 2019)
  - Dynamics of this are governed by denitrification kinetics, which may be affected by varying oxygen availabilities
- **Nitrous oxide (N_2O)**
  - 273x global warming of carbon dioxide (CO_2) (U.S. EPA, 2023)
  - From incomplete denitrification (Davis *et al.*, 2019)
- **Methane (CH_4)**
  - 27-30x global warming of CO_2 (U.S. EPA, 2023)
  - Attributed to methanogenesis, which is associated with complete denitrification (Davis *et al.*, 2019)
- **Poorly characterized in wastewater treatment**

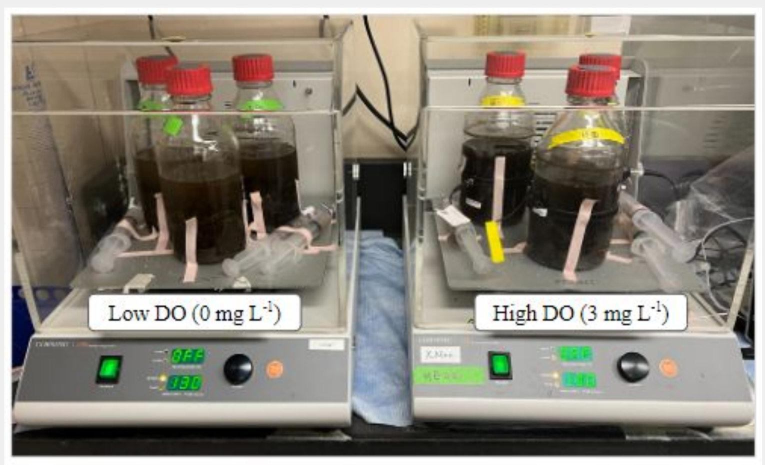
Research Goal

Identify whether lower or higher **DO concentrations** would be optimal in improving **nitrogen removal** while minimizing **greenhouse gas release**

Methods



Experimental setup



Collect samples from **3 trials per group** every **4-8 hours** of **incubation time** for **80 hours;** analyze to determine:

Incubation Time (HRT)

- Incubation time is defined as the time elapsed since the start of data collection
  - Analogized to hydraulic retention time (HRT), the amount of time a given volume of wastewater takes to pass through a bioreactor

Nitrate Removal

- Flow Injection Analysis via Lachat QC8500
- Concentrations of nitrogen species: NH_4^+, NO_3^-, NO_2^-

Greenhouse Gas Production

- Gas Chromatography (GC) via Shimadzu GC-2014
  - The GC separated the sample then detected chemical features
  - Peak areas from the resulting graph were converted to concentrations
- Concentrations of N_2O and CH_4

Global Warming Impact

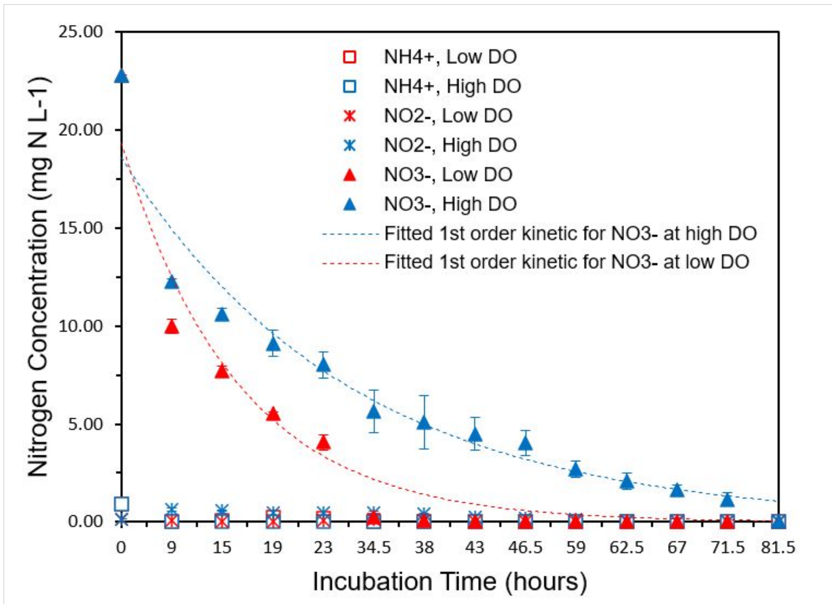
- Greenhouse gas concentrations converted to CO_2 equivalents using global warming potential (Davis et al., 2019)
  - N_2O: 273
  - CH_4: 28

(Davis et al., 2019)

Results

Although complete nitrate removal was observed for both low and high DO concentrations, low DO concentrations resulted in comparatively higher nitrate removal rates

(a)



(b)

$$A_t = A_0 e^{-kt}$$

Group	NO_3^-, Low DO	NO_3^-, High DO
K (hr^{-1})	0.0726	0.0365
A_0 (mg N L^{-1})	21.6325	19.6563
R^2	0.9884	0.9812

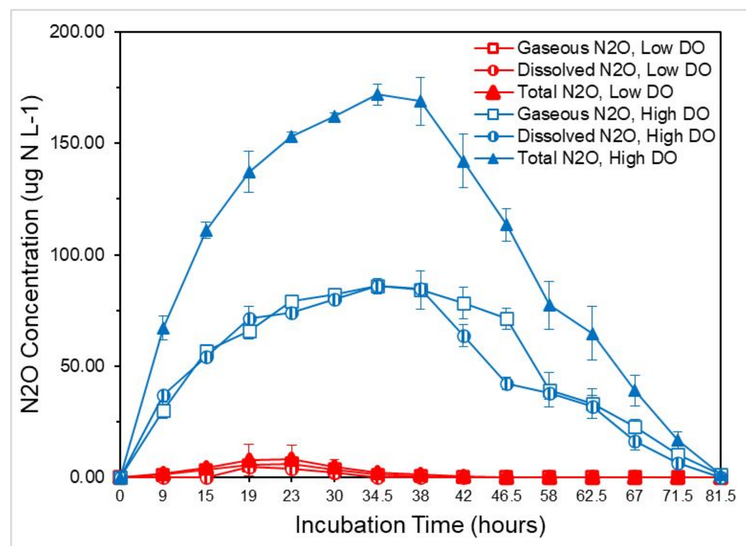
(c)

Avg. NO_3^- removal rate (g N m-3 d-1)	12.7	6.7
Time of complete NO_3^- removal	43 hours	81.5 hours

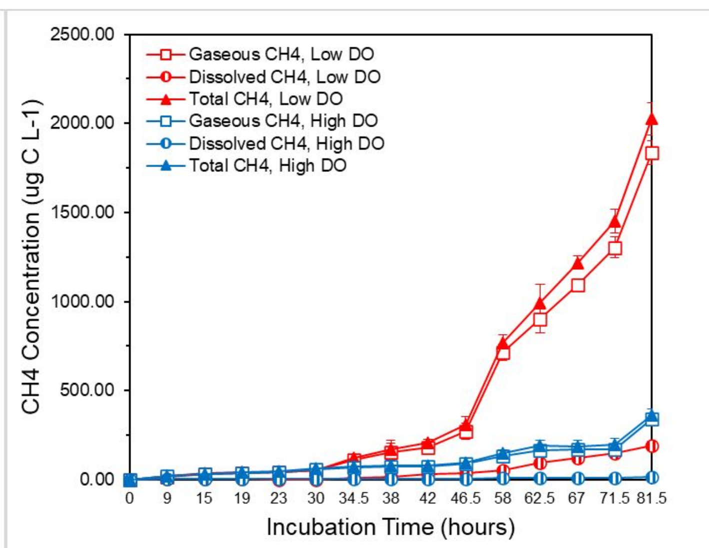
Figure 4. Flow Injection Analysis was used to quantify concentrations of ammonium, nitrite, and nitrate throughout the experiment incubation time of 81.5 hours in order to provide insight on denitrification kinetics. **(a)** Concentrations of nitrogen species over time; error bars represent ±1 SD. **(b)** Parameters of reaction rate equation for denitrification kinetics. **(c)** Average nitrate removal rates prior to complete denitrification and incubation times corresponding to complete nitrate removal. (Image generated by student)

Dependent on initial DO, there is a differential release of greenhouse gases: low DO levels primarily favored methane release, while high DO favored nitrous oxide release

(a)



(b)



(c)

Group	Max total N_2O concentration (μg N L^{-1}) (p < 0.001)		Max total CH_4 concentration (μg C L^{-1}) (p < 0.001)	
Low DO	23 hours	8.31	81.5 hours	2026.64
High DO	34.5 hours	171.97	81.5 hours	363.67

Figure 5. Gas Chromatography was used to quantify concentrations of gaseous, dissolved, and total greenhouse gases. **(a)** N_2O. **(b)** CH_4. **(c)** Maximum greenhouse gas concentrations and corresponding incubation times. Error bars represent ±1 SD. Two-tailed T-tests were used to evaluate statistical significance. (Image generated by student)

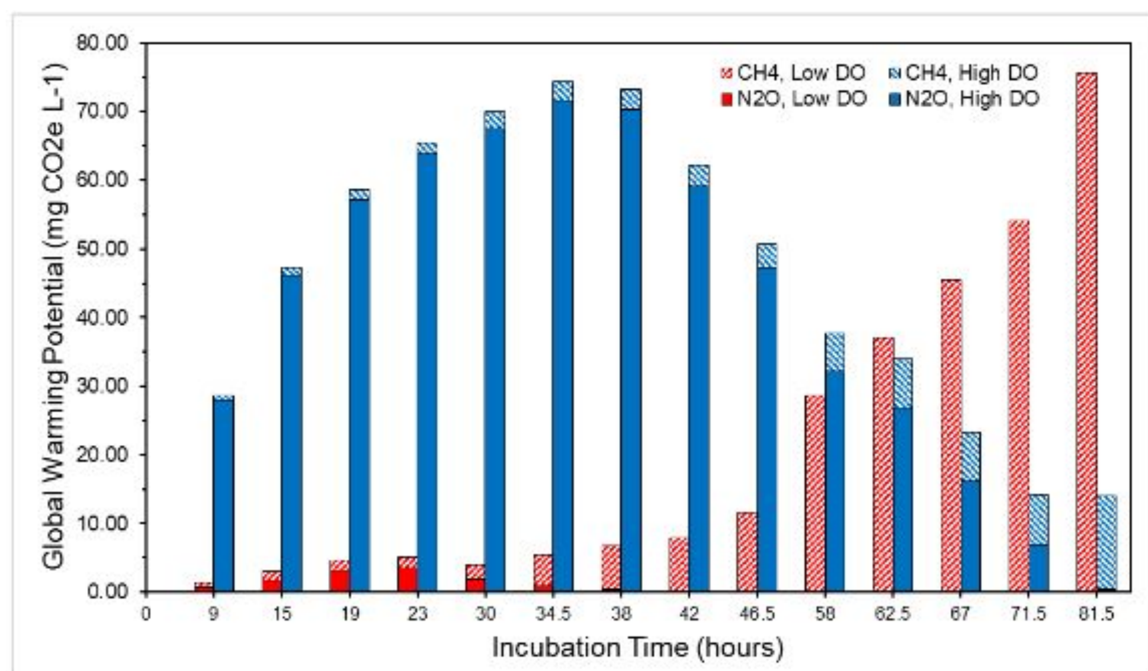


Figure 6. Total (gaseous and dissolved) greenhouse gas concentrations were obtained, stratified (between N2O and CH4), then converted to CO2 equivalents (multiplying by global warming potential for carbon footprint or global warming impact) to be depicted in the column chart for low and high DO groups in the experiment (Davis et al., 2019). (Image generated by student)

Discussion

- **This study aimed to identify whether lower or higher DO concentrations would be optimal in improving nitrogen removal while minimizing greenhouse gas release**
- ***Although both initial DO conditions resulted in complete removal of nitrate, the process was more effective and efficient under low DO conditions (requiring half the time)***
  - Average NO_3^- removal rates similar to literature (Schipper *et al.*, 2010; Kouanda and Hua, 2021).
  - Oxygen inhibits enzymes in denitrification (NO_3^- reductase & N_2O reductase) (Luo *et al.*, 2016)
  - *A higher reaction rate indicates quicker NO_3^- removal; however, its implications on greenhouse gas evolution must be examined*
- ***High DO levels were conducive to high nitrous oxide production due to incomplete denitrification***
  - NO_2^- reductase preferentially receives electrons from available carbon over N_2O reductase (Pan *et al.*, 2013)
  - Sufficient time must be provided for complete denitrification, which minimizes N_2O production (Audet *et al.*, 2021; Rivas *et al.*, 2020)
- ***Low DO levels were conducive to high methane production via methanogenesis***
  - Denitrifying bacteria outcompete methanogens for the same electron donor (H_2); once intermediate denitrification products are reduced, CH_4 production increases (Klüber and Conrad, 1998)
- ***Incubation time (HRT) was found to govern greenhouse gas emissions at both DO levels***
  - Too short → N_2O & incomplete nitrate removal
  - Too long → CH_4 & higher construction costs
  - At correct HRT: low DO → improved nitrate removal and less global warming impact

Limitations

- The pressure within bottles not monitored during experimentation—assumed to be 1 atm
- The composition of nitrified effluent at incubation time of 0 hours was assumed to be uniform across groups
- There was homogenization within the bottles due to use of the shaker (unlike typical NRBs, which are not mixed) (Chen *et al.*, 2022)
- Since the carbon content of the woodchips was not analyzed, there might have been a lack of uniformity among the woodchips

Conclusion

- **Low DO conditions were found to be optimal for denitrifying woodchip bioreactors**
  - 2x quicker complete nitrate removal → less N_2O emissions
  - Correct incubation time → minimized CH_4 emissions

Future Directions

- Develop novel sampling system & model based on results of this study to optimize NRB performance
  - Adjust incubation time based on environmental factors
  - Adjust for changes in temperature (affecting reaction rates)
  - Trap & repurpose methane emissions as natural gas for renewable energy
- Normalize greenhouse gas production by measuring abundance of key functional genes to compare their impact to that of other wastewater treatment systems

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